Chapter 5 — Lists, Tuples and Arrays

Python keeps items **in order** in several ways: the all-purpose `list`, the immutable `tuple`, the compact numeric `array.array`, and NumPy's `ndarray` (NumPy is the standard third-party library for fast numeric arrays). All support indexing, slicing, `len`, `in` and `for`, but they differ in mutability, memory, speed and safety. This chapter explains each, how it works underneath, and how to choose.

What you'll learn

1. The `array` module: type codes, insert/delete/search/update, methods.
2. Lists: creating, accessing, updating, joining, slicing, looping, operators.
3. Lists as dynamic arrays — the big-O cost of every operation.
4. Adding/removing items and the full list-method table.
5. Advanced slicing: steps, slice assignment, slice deletion.
6. List comprehensions.
7. Sorting: `sort()` vs `sorted()`, `key`, `reverse`, stability, multi-key sorts.
8. Copying: aliasing, shallow vs deep, the `[[0]*3]*3` trap.
9. Tuples: immutability, operators, unpacking (with `*`), joining, methods, records, `namedtuple`.
10. Choosing `list`, `tuple`, `array.array` or NumPy.

Runnable code is in `code/`. Outputs are real (Python 3.14; code runs on 3.12+). Timings are from one laptop — ratios hold.

5.1 The `array` module — typed, compact arrays

What it is

An **array** (standard-library `array` module) is a mutable sequence where **every element has the same C numeric type** (a fixed-size C number format) — all 32-bit ints, all 64-bit floats, etc. Values are stored as raw machine numbers side by side in one block (**contiguous** storage), not as separate Python objects. A list is a shelf of boxes holding anything; an array is an egg carton — uniform slots, packed densely.

Why it matters

For large volumes of plain numbers — sensor readings, audio samples, pixel values, binary file formats — an array uses a fraction of a list's memory and reads/writes files and sockets directly as bytes.

How it works

1. Create with a **type code**: `array.array("i", [10, 20, 30])`.
2. Each stored value is converted to that C type; wrong types → `TypeError`, out-of-range → `OverflowError`.
3. Indexing (incl. negative), slicing, `len`, `in`, `for` work as with lists.

Type codes (minimum sizes per the C standard; `itemsize` gives the real size)

Code	C type	Python type	Min bytes	Typical on 64-bit macOS/Linux
`'b'` / `'B'`	signed (allows negatives) / unsigned char	int	1	1 (−128…127 / 0…255)
`'h'` / `'H'`	signed / unsigned short	int	2	2
`'i'` / `'I'`	signed / unsigned int	int	2	4
`'l'` / `'L'`	signed / unsigned long	int	4	8 (4 on Windows)
`'q'` / `'Q'`	signed / unsigned long long	int	8	8
`'f'`	float	float	4	4
`'d'`	double	float	8	8
`'u'`	wchar_t (Unicode char)	str	2	4 — **deprecated** since 3.3, slated for removal
`'w'`	Py_UCS4 (Unicode char)	str	4	4 — added in 3.13 to replace `'u'`

```
a = array.array("i", [10, 20, 30, 40, 50])      # 'i' = C signed int
print(a, a.typecode, a.itemsize, "bytes per item")
print(a[0], a[1], a[-1], a[-2])
for x in a:
    print(x, end=" ")
try:
    a.append(3.5)                               # wrong type is rejected
except TypeError as e:
    print("TypeError:", e)
try:
    array.array("b", [200])                     # 'b' holds -128..127
except OverflowError as e:
    print("OverflowError:", e)
```

Output:

```
array('i', [10, 20, 30, 40, 50]) i 4 bytes per item
10 20 50 40
10 20 30 40 50
TypeError: 'float' object cannot be interpreted as an integer
OverflowError: signed char is greater than maximum
```

The four basic operations: insert, delete, search, update

Key term — Big-O notation: how work grows with *n* elements. **O(1)**: constant. **O(n)**: linear. **O(k)**: linear in *k*, the size of the extra input. Growth, not seconds.

Insert (`insert` , shifts the rest right), **delete** (`pop(i)` / `remove(x)` , shift left), **search** (`index`) and **traverse** (`for x in a`) are O(n); **update** (`a[i] = x`) is O(1) — see the method table.

Array methods

Method	What it does	Cost
`append(x)`	Add at the end	O(1) amortised (constant on average)
`extend(iterable)`	Append every item (same type required)	O(k)
`insert(i, x)`	Insert before `i`	O(n)
`pop([i])`	Remove and return item `i` (default last)	O(1) last, O(n) otherwise
`remove(x)`	Delete first `x`	O(n)
`index(x)`	Position of first `x`	O(n)
`count(x)`	Occurrences of `x`	O(n)
`reverse()`	Reverse in place	O(n)
`clear()`	Remove all (**3.13+**; older: `del a[:]`)	O(1)
`tolist()` / `fromlist(l)`	To / append from a list	O(n)
`tobytes()` / `frombytes(b)`	Raw machine bytes out / in	O(n)
`tofile(f)` / `fromfile(f, n)`	Write / read raw values to a binary file	O(n)
`byteswap()`	Swap byte order (endianness) of every item	O(n)
`buffer_info()`	`(address, length)` of the underlying memory	O(1)

Arrays have **no** `.sort()` or `.copy()`. Sort with `sorted(a)` (returns a **list**) and rebuild with `array.array(code, sorted(a))`; copy with `a[:]`, `array.array(a.typecode, a)` or `copy.copy(a)`.

```
a = array.array("i", [1, 2, 3, 4, 5])
a.insert(1, 10);            print("insert(1, 10):", a.tolist())
removed = a.pop(1);         print("pop(1):", a.tolist(), "removed", removed)
print("index(3):", a.index(3))
a[2] = 20;                  print("a[2] = 20:", a.tolist())
a.append(4);                print("append(4):", a.tolist())
a.extend([5, 4, 6]);        print("extend:", a.tolist())
print("count(4):", a.count(4))
a.remove(4);                print("remove(4) (first 4 only):", a.tolist())
a.reverse();                print("reverse:", a.tolist())
print("sorted(a) -> list:", sorted(a))          # arrays have no .sort()
a = array.array("i", sorted(a));  print("back to array:", a)
b = copy.copy(a)                                 # no .copy() method: use copy.copy or a[:]
b[0] = 99
print("copy independent:", a[0], b[0])
del a[:]                                         # empties it (array.clear() exists from 3.13)
print("emptied:", a)
```

Output:

```
insert(1, 10): [1, 10, 2, 3, 4, 5]
pop(1): [1, 2, 3, 4, 5] removed 10
index(3): 2
a[2] = 20: [1, 2, 20, 4, 5]
append(4): [1, 2, 20, 4, 5, 4]
extend: [1, 2, 20, 4, 5, 4, 5, 4, 6]
count(4): 3
remove(4) (first 4 only): [1, 2, 20, 5, 4, 5, 4, 6]
reverse: [6, 4, 5, 4, 5, 20, 2, 1]
sorted(a) -> list: [1, 2, 4, 4, 5, 5, 6, 20]
back to array: array('i', [1, 2, 4, 4, 5, 5, 6, 20])
copy independent: 1 99
emptied: array('i')
```

Memory. One million integers:

```
list of 1,000,000 ints :   36.0 MB (pointers + int objects)
array('i')             :    4.1 MB (raw 4-byte ints)
```

A list stores an 8-byte pointer (memory address) per slot **plus** a separate int object (28 bytes for small ints on 64-bit CPython) — ~9× more here.

Trade-offs, mistakes, and when not to use it

- **Not faster for arithmetic.** Every read creates an `int` / `float` object ("boxing"), so looping over an array is often *slower* than a list (5.14). For fast maths use NumPy.
- **Choose the type code carefully:** `'b'` overflows at 128; `'l'` is 4 bytes on Windows, 8 on Linux/macOS — use `'q'` for 64 bits everywhere.
- Arrays are for **compact storage and binary I/O**; lists for general work; NumPy for numerics.

TL;DR

`array.array(code, data)` = homogeneous, compact, mutable numbers; list-like indexing and most methods; no `sort` / `copy`; `clear()` from 3.13. A memory win, not a speed win.

5.2 Lists — the everyday sequence

What it is

A **list** is an ordered, **mutable** collection of objects of **any type**, written `[1, "Hello", 3.14, True]`, that grows and shrinks at run time. It's Python's default container: function results, file rows, API items, stacks (last-in-first-out) and queues (first-in-first-out) of work.

How it works — create, access, update

```python
my_list = [1, "Hello", 3.14, True]          # mixed types are fine
print(my_list, len(my_list))
print(my_list[0], my_list[1], my_list[-1], my_list[-2])
my_list[1] = "World"
print("single update:", my_list)
my_list[1:3] = ["Changed", "Completely"]
print("slice update:", my_list)
print(list("abc"), list(range(4)), [0] * 3, [])
```

Output:

```
[1, 'Hello', 3.14, True] 4
1 Hello True 3.14
single update: [1, 'World', 3.14, True]
slice update: [1, 'Changed', 'Completely', True]
['a', 'b', 'c'] [0, 1, 2, 3] [0, 0, 0] []
```

Indexing works like strings; unlike strings you can **assign** to an index or slice (5.5). `list(iterable)` builds a list from anything iterable.

How it works — joining, slicing, looping

```
a, b = [1, 2, 3], [4, 5, 6]
c = a + b                                    # new list
a.extend(b)                                  # modifies a in place
print(c, a, c == a, c is a)
full = list(range(10))
print(full[2:7], full[-8:-3], full[::3], full[::-1])
for i, fruit in enumerate(["apple", "banana", "cherry"], start=1):
    print(i, fruit)
```

Output:

```
[1, 2, 3, 4, 5, 6] [1, 2, 3, 4, 5, 6] True False
[2, 3, 4, 5, 6] [2, 3, 4, 5, 6] [0, 3, 6, 9] [9, 8, 7, 6, 5, 4, 3, 2, 1, 0]
1 apple
2 banana
3 cherry
```

`+` builds a **new** list; `extend()` / `+=` **modify** the left list. `==` compares contents; `is` identity.

List operators

Operator	Meaning	Example	Result
`+`	Concatenate into a new list	`[1, 2] + [3]`	`[1, 2, 3]`
`*`	Repeat	`[1, 2] * 2`	`[1, 2, 1, 2]`
`in` / `not in`	Membership (linear scan)	`3 in [1, 2, 3]`	`True`
`==`, `<`, …	Element-by-element (lexicographic)	`[1, 2, 3] < [1, 3]`	`True`
`+=`	Extend in place	`a += [4]`	`a` changed
`del a[i]`	Delete by index or slice	`del a[0]`	

```
a, b = [1, 2, 3], [4, 5, 6]
print(a + b, a * 3, 3 in a, 7 not in a)
print([1, 2, 3] < [1, 3], [1, 2] == [1, 2], max(a), min(a), sum(a))
```

Output:

```
[1, 2, 3, 4, 5, 6] [1, 2, 3, 1, 2, 3, 1, 2, 3] True True
True True 3 1 6
```

`[1, 2, 3] < [1, 3]`: first elements tie, then 2 < 3 — like comparing words.

Common mistakes

- `a = a.extend(b)` sets `a` to `None` (`extend` returns `None`).
- Repeated `in` on a big list inside a loop — O(n) each; convert to a `set` (Chapter 6).
- Removing while iterating skips elements; iterate a copy (`lst[:]`) or rebuild with a comprehension.

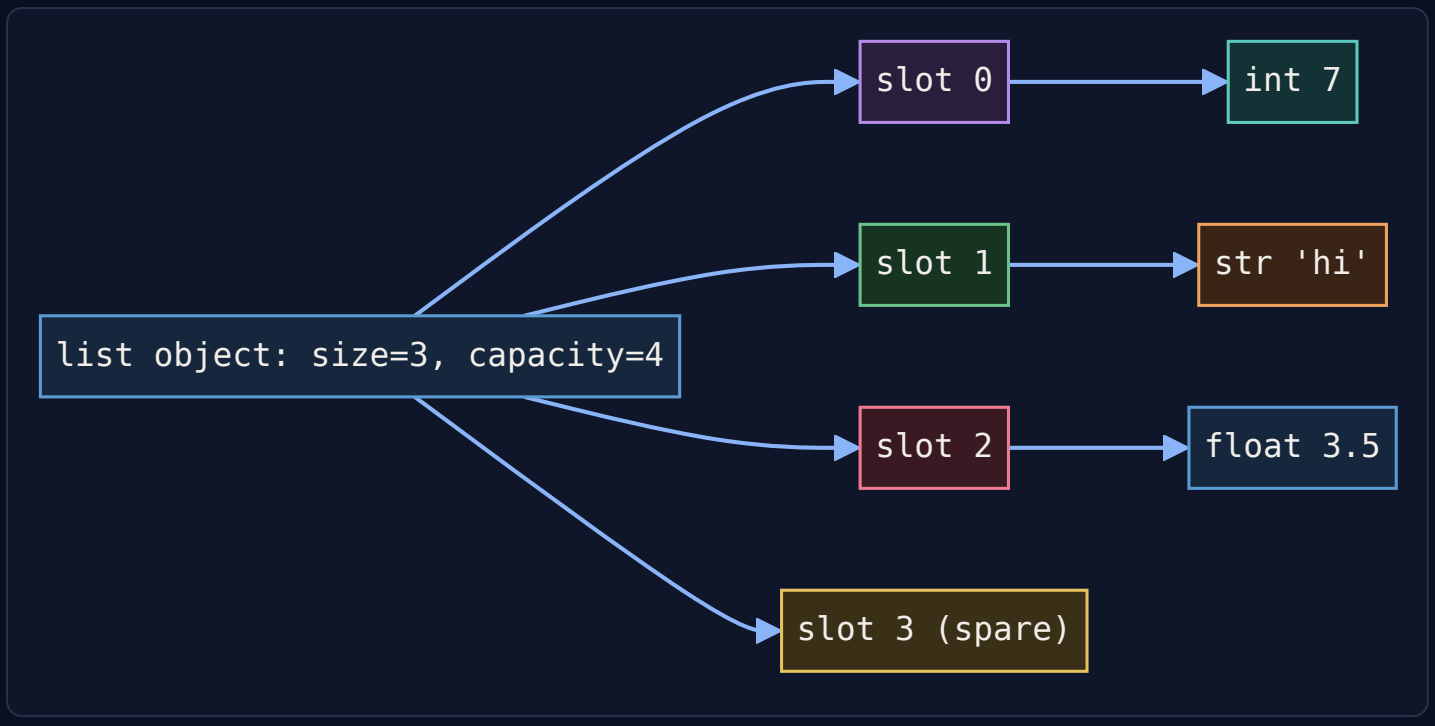
TL;DR

Ordered, mutable, heterogeneous (mixed types). Index, slice, update in place; `+` makes new lists, `extend` / `+=` modify.

5.3 List internals — a dynamic array — and the big-O table

What it is

In CPython a list is a **dynamic array**: one contiguous block of **pointers** to element objects, plus the used length (`len`) and the allocated slots (**capacity**). "Dynamic": it resizes automatically. The layout tells you which operations are cheap — the difference between O(n) and O(n²) programs.



How it works

1. **Index** `lst[i]`: address = start + i × 8 bytes → O(1).
2. **Append** with spare capacity: write the next slot, O(1).
3. **Append when full**: allocate a bigger block, copy all pointers, write. Growth is **~12.5% plus a small constant** (recent CPython: `new_size + new_size/8 + 6`, rounded down to a multiple of 4). Each resize leaves room for many appends, so the copying averages to a constant — **amortised O(1)**.
4. **Insert/delete at i**: every later pointer shifts → O(n − i); at the front, the whole list.
5. **Shrinking**: below half capacity, CPython may reallocate smaller.

Measured capacity while appending (`code/06_list_internals.py` , via `sys.getsizeof`):

```
len  capacity  (capacity = slots allocated)
  0         0
  1         4
  5         8
  9        16
 17        24
 25        32
 33        40
 41        52
 53        64
 65        76
```

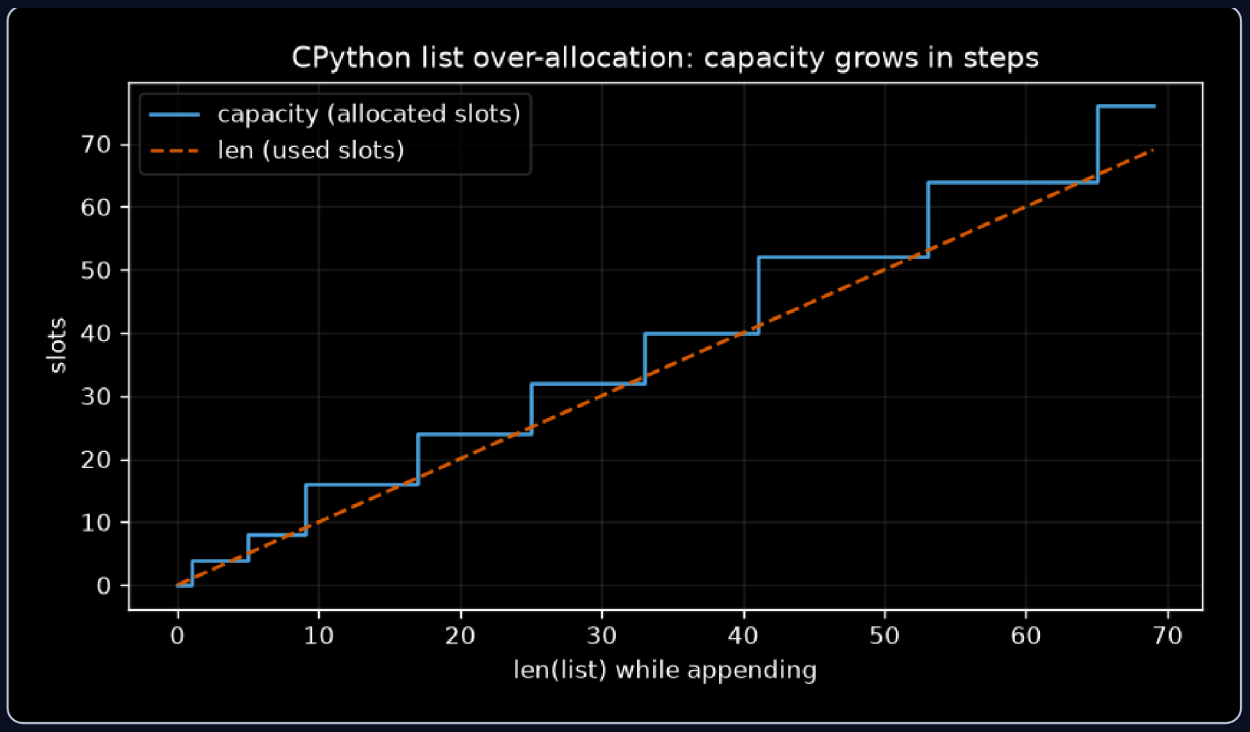


Figure — capacity (steps) stays at or above the used length (dashed); a copy happens only at each step.

Worked example: the 17th append finds 16 full, allocates 24 and copies 16 pointers; the next 7 appends are free — 2 copies each over 8 appends. Steps grow with the list, so the average stays bounded.

End vs front (µs — millionths of a second — per push-then-pop pair):

n	append()+pop()	insert(0)+pop(0)	(µs per pair)
10,000	0.017	3.588	
100,000	0.017	40.046	
1,000,000	0.018	414.395	

End operations stay flat as *n* grows ×100; front operations grow ×100 — O(1) vs O(n).

Big-O table for `list` (n = len, k = size of the other argument or slice)

Operation	Example	Average cost
Length	`len(a)`	O(1)
Get / set item	`a[i]`, `a[i] = x`	O(1)
Append	`a.append(x)`	O(1) amortised
Pop last	`a.pop()`	O(1)
Pop / delete at i	`a.pop(i)`, `del a[i]`	O(n − i) → O(n) at the front
Insert at i	`a.insert(i, x)`	O(n − i)
Extend	`a.extend(b)`, `a += b`	O(k)
Concatenate	`a + b`	O(n + k)
Repeat	`a * k`	O(n·k)
Get slice	`a[i:j]`	O(k)
Set slice	`a[i:j] = b`	O(k + n)
Delete slice	`del a[i:j]`	O(n)
Membership / search	`x in a`, `a.index(x)`, `a.count(x)`, `a.remove(x)`	O(n)
Min / max / sum	`min(a)`	O(n)
Copy	`a.copy()`, `a[:]`	O(n)
Reverse	`a.reverse()`	O(n)
Sort	`a.sort()`, `sorted(a)`	O(n log n) worst (slightly more than linear — the norm for good sorts), O(n) if already sorted
Iterate	`for x in a`	O(n)

Trade-offs

Need	Best structure	Why
Append/pop at the end (stack)	`list`	O(1)
Add/remove at both ends (queue)	`collections.deque` (double-ended queue; Chapter 6)	O(1) at both ends; `list.pop(0)` is O(n)
Frequent membership tests	`set` / `dict`	O(1) average vs O(n)
Keep sorted while inserting	`bisect` (binary-search helpers) on a list (Chapter 6) or a sorted-container library	O(log n) search (doubling n adds one step), O(n) insert

Common mistakes

- `list.pop(0)` as a queue — quadratic for large queues; use `deque.popleft()`.
- `insert(0, x)` in a loop to build a reversed list — append, then `reverse()` once.

In the interview

30-second answer: "A Python list is a dynamic array of pointers. Indexing and append are O(1) (append amortised, thanks to proportional over-allocation of about 12.5%), but inserting or

deleting anywhere except the end shifts elements and is O(n), and membership tests are O(n)."

- *Why amortised?* → Rare resizes copy everything, but total copying over n appends is O(n).
- *Linked list vs Python list?* → No built-in linked list (a chain of nodes each pointing to the next); `deque` is a doubly-linked list (nodes point both ways) of fixed-size blocks: O(1) at both ends, O(n) indexing in the middle.
- *Why is `x in lst` slow?* → Linear scan.
- *What's in the slots?* → Pointers, not the objects.

TL;DR

Dynamic array of pointers. O(1): index, append, pop-last. O(n): insert/delete at front or middle, `in`, `index`, `remove`. Queue → `deque`; lookups → `set`.

5.4 Adding and removing items

What it is

Goal	Use	Returns	Error when
Add one item at the end	`append(x)`	`None`	—
Add one item at position i	`insert(i, x)`	`None`	— (i beyond ends is clamped to the nearest end)
Add all items of an iterable	`extend(iterable)` / `+=`	`None`	Not iterable
Remove first item equal to x	`remove(x)`	`None`	`ValueError` if absent
Remove by position and get it	`pop(i=-1)`	The item	`IndexError` on empty/out of range
Remove by position or slice	`del a[i]`, `del a[i:j]`	—	`IndexError`
Remove everything	`clear()` (or `del a[:]`)	`None`	—

```
fruits = ["apple", "banana"]
fruits.append("cherry");              print("append:", fruits)
fruits.insert(1, "orange");           print("insert:", fruits)
fruits.extend(["grape", "mango"]);    print("extend:", fruits)
fruits.remove("banana");              print("remove:", fruits)
last = fruits.pop();                  print("pop():", fruits, "->", last)
first = fruits.pop(0);                print("pop(0):", fruits, "->", first)
del fruits[0];                        print("del [0]:", fruits)
fruits.clear();                       print("clear:", fruits)
x = [1, 2]
x.append([3, 4]);  print("append a list :", x)   # nests!
y = [1, 2]
y.extend([3, 4]);  print("extend a list :", y)
try:
    y.remove(99)
except ValueError as e:
    print("ValueError:", e)
```

Output:

```
append: ['apple', 'banana', 'cherry']
insert: ['apple', 'orange', 'banana', 'cherry']
extend: ['apple', 'orange', 'banana', 'cherry', 'grape', 'mango']
remove: ['apple', 'orange', 'cherry', 'grape', 'mango']
pop(): ['apple', 'orange', 'cherry', 'grape'] -> mango
pop(0): ['orange', 'cherry', 'grape'] -> apple
del [0]: ['cherry', 'grape']
clear: []
append a list : [1, 2, [3, 4]]
extend a list : [1, 2, 3, 4]
ValueError: list.remove(x): x not in list
```

Common mistakes

- `append([3, 4])` adds **one** (nested) element; `extend([3, 4])` adds **two**; `extend("hi")` adds `'h'`, `'i'`.
- `while x in lst: lst.remove(x)` — O(n²). Use `lst = [v for v in lst if v != x]`.
- `lst.clear()` empties the **same** object (all references see it); `lst = []` rebinds only this name.

TL;DR

`append` one, `extend` many, `insert` at a position; `remove` by value (first only), `pop` by index (returns it), `del` by index/slice, `clear` everything.

5.5 List methods (complete table) and slicing in depth

The full method table

Method	Does	Cost
`append(x)`	Add at the end	O(1) amortised
`extend(it)`	Add every item of `it`	O(k)
`insert(i, x)`	Insert before `i`	O(n)
`remove(x)`	Delete first equal item	O(n)
`pop([i])`	Remove and return item `i` (default last)	O(1) last / O(n)
`clear()`	Remove all	O(n) (drops each reference)
`index(x[, start[, end]])`	First index of `x`	O(n)
`count(x)`	Items equal to `x`	O(n)
`sort(*, key=None, reverse=False)`	Sort in place (stable — equal items keep their order, §5.7)	O(n log n)
`reverse()`	Reverse in place	O(n)
`copy()`	Shallow copy	O(n)

Only 11 methods; everything else is a built-in (`len`, `sorted`, `reversed`, `min`, `max`, `sum`, `enumerate`, `zip`) or an operator.

```
colors = ["red", "blue", "green", "blue"]
print(colors.count("blue"), colors.index("blue"), colors.index("blue", 2))
colors.remove("blue");   print(colors)          # only the first 'blue'
colors.append("amber")
colors.sort();           print("sort:", colors)
colors.reverse();        print("reverse:", colors)
nums = [3, 1, 2]
snapshot = nums.copy()
nums.sort(reverse=True)
print(nums, snapshot)
```

Output:

```
2 1 3
['red', 'green', 'blue']
sort: ['amber', 'blue', 'green', 'red']
reverse: ['red', 'green', 'blue', 'amber']
[3, 2, 1] [3, 1, 2]
```

`index("blue", 2)` starts at position 2, finding the second `"blue"` at 3.

Slicing in depth — steps and slice assignment

String slicing rules (Chapter 4) apply. Because lists are mutable you can also **assign** and **delete** slices.

Form	Rule
`a[i:j] = iterable`	Replace that range; new data may be **longer or shorter**
`a[i:i] = iterable`	Empty range → pure **insertion** at `i`
`a[i:j:k] = iterable`	Extended (stepped) slice → lengths **must match**
`del a[i:j:k]`	Delete the selected positions
`a[:] = []`	Empty in place (same as `clear()`)

```
x = list(range(10))
print(x[1:8:2], x[::-2], x[-3:])
y = x.copy(); y[2:5] = ["a"];           print("shrink :", y)      # lengths may differ
y = x.copy(); y[2:2] = ["new", "items"]; print("insert :", y)      # empty slice = insert
y = x.copy(); y[::2] = [0] * 5;         print("stepped:", y)      # lengths must match
y = x.copy(); del y[::2];               print("del ::2:", y)
y = x.copy(); y[:] = [];                print("clear  :", y)
try:
    y = x.copy(); y[::2] = [1, 2]
except ValueError as e:
    print("ValueError:", e)
```

Output:

```
[1, 3, 5, 7] [9, 7, 5, 3, 1] [7, 8, 9]
shrink : [0, 1, 'a', 5, 6, 7, 8, 9]
insert : [0, 1, 'new', 'items', 2, 3, 4, 5, 6, 7, 8, 9]
stepped: [0, 1, 0, 3, 0, 5, 0, 7, 0, 9]
del ::2: [1, 3, 5, 7, 9]
clear  : []
ValueError: attempt to assign sequence of size 2 to extended slice of size 5
```

Common mistakes

- `a[1:3] = 5` — the right side must be iterable: `a[1:3] = [5]`.
- A slice **read** is a shallow copy; a slice **assignment** changes the original.

TL;DR

11 methods; slicing reads copies, slice assignment edits in place (length may change unless there's a step).

5.6 List comprehensions

What it is

A **list comprehension** builds a new list in one expression:

```
[expression for item in iterable if condition]
```

- **expression** — what goes in (e.g. `n ** 2`).
- **for item in iterable** — the source loop.
- **if condition** — optional filter.

The problem it solves

Empty-list + loop + `append` takes three or four lines to say one thing, and the repeated `.append` attribute lookup and call cost time. A comprehension is one readable line compiled to specialised bytecode.

How it works

1. Create an empty list; for each item, test `condition` (if any).
2. If true, evaluate `expression` and append.
3. Multiple `for` clauses nest **left to right**, as nested loops would.

```python
numbers = [1, 2, 3, 4, 5]
squared = []
for n in numbers:                          # the traditional way
    squared.append(n ** 2)
print("loop:", squared)
print("comp:", [n ** 2 for n in numbers])
print("even squares:", [n ** 2 for n in numbers if n % 2 == 0])
print("if/else in expr:", ["even" if n % 2 == 0 else "odd" for n in numbers])
print([f"Number {n} squared is {n ** 2}." for n in numbers[:2]])

matrix = [[1, 2, 3], [4, 5, 6]]
flat = [x for row in matrix for x in row]          # same order as nested for-loops
transposed = [[row[i] for row in matrix] for i in range(3)]
pairs = [(c, n) for c in "ab" for n in (1, 2)]
print(flat, transposed, pairs)

total = sum(n * n for n in range(1_000_000))       # generator expression: no list built
```

Output:

```
loop: [1, 4, 9, 16, 25]
comp: [1, 4, 9, 16, 25]
even squares: [4, 16]
if/else in expr: ['odd', 'even', 'odd', 'even', 'odd']
['Number 1 squared is 1.', 'Number 2 squared is 4.']
[1, 2, 3, 4, 5, 6] [[1, 4], [2, 5], [3, 6]] [('a', 1), ('a', 2), ('b', 1), ('b', 2)]
sum of squares: 333332833333500000
loop+append: 39.0 ms   comprehension: 31.6 ms (ratio 1.2x; machine-dependent)
```

Two `if` positions:

- `[... for n in xs if cond]` — **filter** (fewer items).
- `[a if cond else b for n in xs]` — **choose a value** (same count).

The speed gain is modest (1.2×); choose comprehensions for **clarity**.

Trade-offs

	Comprehension	for + append	Generator expression (…) (lazy, one item at a time)	map / filter (apply a function / keep matches)
Readability for simple transforms	Best	OK	Best	OK with named functions
Memory	Whole list	Whole list	One item at a time	Lazy iterator
Complex logic, try/except, prints	Poor	Best	Poor	Poor

When not to use it

- Several statements, error handling, side effects (printing, writing files) — write a loop.
- One-pass consumers (`sum` , `any` , `max` , `"".join`) — a **generator expression** avoids materialising (building in memory) the list: `sum(x*x for x in xs)` .
- More than two `for` clauses or nesting deeper than one level (*Effective Python*: at most two control subexpressions).
- Side effects only (`[print(x) for x in xs]`) builds a useless list of `None` s.

Common mistakes

- Wrong nesting order: `[x for x in row for row in matrix]` fails (`row` undefined).
- Expecting the loop variable to leak: in Python 3 it's local to the comprehension.

Real-world examples

Data cleaning (`[c.strip().lower() for c in columns]`), SQL parameter lists, flattening paged API results (`[item for page in pages for item in page["results"]]`) in ETL (extract–transform–load) scripts.

In the interview

> **30-second answer:** " `[expr for x in it if cond]` builds a list in one expression; it's clearer and a bit faster than loop-and-append. Use a generator expression when you only need to iterate once, and a normal loop when the logic gets complicated."

- *Order of multiple fors?* → Left to right, like nested loops.
- *Dict and set versions?* → `{k: v for ...}` , `{x for ...}` (Chapter 6).

Self-check

1. `[c.upper() for c in "abc" if c != "b"]` ? → `['A', 'C']` .
2. Flatten `[[1],[2,3]]` ? → `[x for sub in lst for x in sub]` .

TL;DR

`[expr for x in it if cond]` ; `if/else` goes before `for` ; nested `for` s read left to right; `( … )` for one-pass use; loops for complex logic.

5.7 Sorting

What it is

- `lst.sort()` — sorts **in place**, returns `None` .
- `sorted(iterable)` — returns a **new** list from any iterable (list, tuple, string, set, dict keys, generator).

Both take `key=` (function computing the sort value) and `reverse=True` . Sorting underlies reports, rankings, deduplication, binary search (repeatedly halving a sorted list) and top-k queries; `key` plus

stability express almost any ordering in one line.

How it works

1. `key` is called **once per element**; Python sorts by the results (decorate–sort–undecorate for you).
2. Items (or keys) are compared with `<`; incomparable mixed types raise `TypeError`.
3. Algorithm: **Timsort** (since 2.3; 3.11 adopted the "powersort" merge policy). It finds runs (stretches already in order) and merges them: **O(n log n)** worst, **O(n)** on sorted data.
4. **Stable**: equal keys keep their original order — so multi-key sorts can be done in passes, secondary key first.

```
nums = [5, 2, 9, 1]
new = sorted(nums)                  # returns a NEW list, works on any iterable
print(new, nums)
result = nums.sort()                # sorts IN PLACE, returns None
print(nums, result)
print(sorted("banana"), sorted({3, 1, 2}), sorted({"b": 1, "a": 2}))

words = ["banana", "Apple", "cherry", "Date"]
print(sorted(words))                                # uppercase first (code points)
print(sorted(words, key=str.casefold))              # case-insensitive
print(sorted(words, key=len))                       # by length
print(sorted(words, key=len, reverse=True))
people = [("Ann", 31), ("Bob", 25), ("Cid", 31)]
print(sorted(people, key=itemgetter(1)))            # same as key=lambda p: p[1]
```

Output:

```
[1, 2, 5, 9] [5, 2, 9, 1]
[1, 2, 5, 9] None
['a', 'a', 'a', 'b', 'n', 'n'] [1, 2, 3] ['a', 'b']
['Apple', 'Date', 'banana', 'cherry']
['Apple', 'banana', 'cherry', 'Date']
['Date', 'Apple', 'banana', 'cherry']
['banana', 'cherry', 'Apple', 'Date']
[('Bob', 25), ('Ann', 31), ('Cid', 31)]
```

Stability: with `key=len`, `"banana"` and `"cherry"` (both 6) keep their order — even with `reverse=True`, which flips comparisons, not tie order.

Multi-key sorts — department ascending, then age descending:

```
people = [("Ann", "HR", 31), ("Bob", "IT", 25), ("Cid", "HR", 25), ("Dee", "IT", 31)]
by_age = sorted(people, key=itemgetter(2))
print("by age (ties keep original order):", [p[0] for p in by_age])
print("tuple key :", [p[0] for p in sorted(people, key=lambda p: (p[1], -p[2]))])
tmp = sorted(people, key=itemgetter(2), reverse=True)       # secondary key first
print("two passes:", [p[0] for p in sorted(tmp, key=itemgetter(1))])
```

Output:

```
by age (ties keep original order): ['Bob', 'Cid', 'Ann', 'Dee']
tuple key : ['Ann', 'Cid', 'Dee', 'Bob']
two passes: ['Ann', 'Cid', 'Dee', 'Bob']
```

A **tuple key** compares field by field; negating a number flips its direction. If a field can't be negated (a string to sort descending), use two stable passes.

Trade-offs

	`list.sort()`	`sorted()`
Input	Lists only	Any iterable
Result	Same list, changed	New list
Memory	No extra list	Extra O(n)
Keep original?	No	Yes

Common mistakes

- `lst = lst.sort()` → `None`.
- Mixed types (`[3, "a"]`) → `TypeError: '<' not supported between instances of 'str' and 'int'`.
- Numeric strings sorted as strings (`"10" < "9"`) — `key=int`.
- A `lambda` (one-line anonymous function) where `operator.itemgetter` / `attrgetter` (ready-made key functions) or `str.casefold` exist — clearer and slightly faster.

Real-world examples

Timsort (Tim Peters, for CPython) was adopted for object sorting in Java 7 (`Arrays.sort`), Android, and V8 (Chrome/Node.js JavaScript engine) since 2018. Leaderboards, "price then rating" filters, SQL `ORDER BY a, b DESC` and pandas `df.sort_values(["a", "b"], ascending=[True, False])` are key-based stable sorts.

In the interview

> **30-second answer:** "`sort` sorts a list in place and returns None; `sorted` returns a new list from any iterable. Both take `key` and `reverse`. Python uses Timsort — stable, O(n log n), O(n) on nearly sorted data — and stability lets you do multi-key sorts with a tuple key or successive passes."

- *Why is stability useful?* → Equal elements keep prior order, enabling multi-pass sorts.
- *How often is `key` called?* → Once per element.
- *Top 3 of a million?* → `heapq.nlargest(3, items)` (a *heap* keeps the smallest on top) — O(n log k) (Chapter 6).

TL;DR

`sort()` in place → `None`; `sorted()` new list; `key=` once per item; Timsort is stable and adaptive.

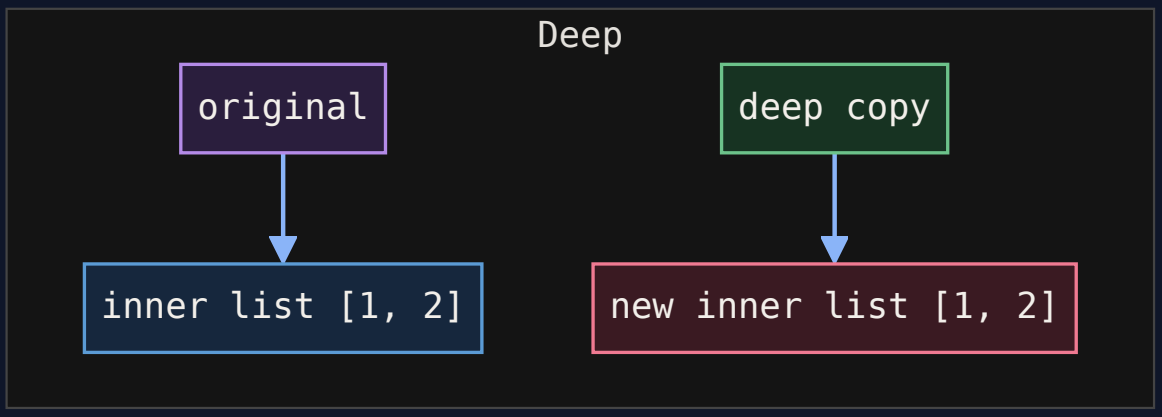
5.8 Copying — aliasing, shallow vs deep, and the nested-list trap

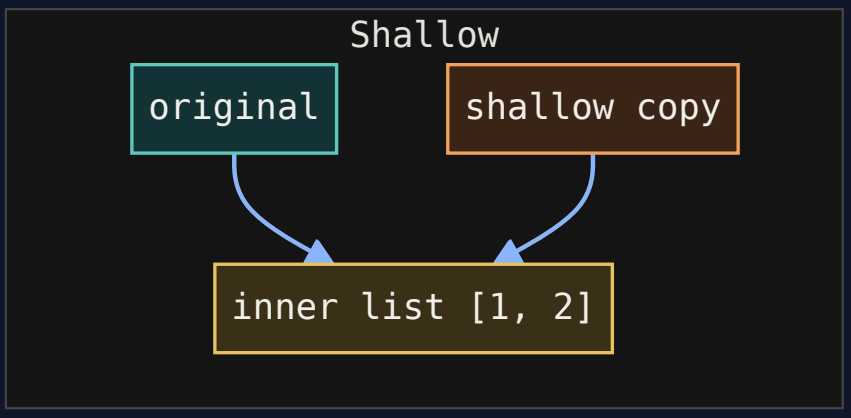
What it is

- **Aliasing**: two names, **same** object. `b = a` never copies.
- **Shallow copy**: new outer container, slots pointing to the **same** inner objects — `a.copy()`, `list(a)`, `a[:]`, `copy.copy(a)`.
- **Deep copy**: new outer container **and** recursive copies of everything inside — `copy.deepcopy(a)`.

Most "my data changed by itself" bugs come from aliasing or shallow copies of nested data.

How it works






```
a = [1, 2, 3]
b = a                        # NOT a copy: two names, one list
b.append(4)
print(a, b, a is b)

original = [[1, 2], [3, 4]]
shallow = original.copy()
deep = copy.deepcopy(original)
original[0].append(99)                  # mutate an INNER list
original.append([5])                    # mutate the OUTER list
print("original:", original)
print("shallow :", shallow, "<- shares inner lists")
print("deep    :", deep, "<- fully independent")
print(shallow[0] is original[0], deep[0] is original[0])
```

Output:

```
[1, 2, 3, 4] [1, 2, 3, 4] True
original: [[1, 2, 99], [3, 4], [5]]
shallow : [[1, 2, 99], [3, 4]] <- shares inner lists
deep    : [[1, 2], [3, 4]] <- fully independent
True False
```

The shallow copy didn't get `[5]` (separate outer list) but saw `99` (shared inner list). Passing a list to a function passes a reference too — `add_item(a)` appending `'x'` gives `[1, 2, 3, 4, 'x']` in the caller.

The `[[0] * 3] * 3` trap

`*` repeats **references**: one inner row, three pointers.

```
bad = [[0] * 3] * 3                    # 3 references to ONE row
bad[0][0] = 1
good = [[0] * 3 for _ in range(3)]     # 3 separate rows
good[0][0] = 1
```

Output:

```
bad : [[1, 0, 0], [1, 0, 0], [1, 0, 0]] ids: 1 distinct row(s)
good: [[1, 0, 0], [0, 0, 0], [0, 0, 0]] ids: 3 distinct row(s)
```

(`[0] * 3` is fine: ints are immutable.)

The mutable default argument trap

Defaults are evaluated **once**, when `def` runs, so a list default is shared by every call:

```
def bad_append(item, bucket=[]):
    bucket.append(item); return bucket
def good_append(item, bucket=None):
    if bucket is None:
        bucket = []
    bucket.append(item); return bucket
print(bad_append(1), bad_append(2))    # [1, 2] [1, 2]
print(good_append(1), good_append(2))  # [1] [2]
```

Trade-offs

	Alias `b = a`	Shallow `a.copy()`	Deep `copy.deepcopy(a)`
Cost	O(1)	O(n)	O(total size of everything)
Independent outer list	No	Yes	Yes
Independent nested objects	No	No	Yes
Handles cycles (objects that contain each other)	—	—	Yes (memo of already-copied objects)

When not to use it

Don't `deepcopy` by reflex — it's slow and copies things you may want shared (caches, database connections). For flat or immutable-only data a shallow copy is effectively full. Immutable structures (tuples, frozen dataclasses — Chapter 10) avoid copying altogether.

Common mistakes

- `grid = [[None] * cols] * rows` — use a comprehension.
- Thinking `a[:]` copies nested lists.
- `def f(x=[])` / `def f(d={})` — use `None` and create inside.

In the interview

30-second answer: "Assignment never copies — it creates another reference. A shallow copy (`copy()` , `list()` , `[:]`) duplicates only the outer container; nested objects are shared. `copy.deepcopy` copies recursively. Classic traps: `[[0]*3]*3` shares one row, and mutable default arguments are shared between calls."

- *Does `a[:]` copy?* → Shallowly.
- *Does deepcopy handle cycles?* → Yes, via a memo dict.

Self-check

1. `a = [[1]]; b = a[:]; b[0].append(2)` — `a`? → `[[1, 2]]`.

TL;DR

`=` aliases, `copy()` shares the insides, `deepcopy` shares nothing. Never multiply a list of lists.

5.9 Tuples — immutable sequences and records

What it is

A **tuple** is an ordered, **immutable** sequence. **Commas** make it; parentheses only group: `(1, 2, 3)` or `1, 2, 3` (**tuple packing**). One element needs a trailing comma: `(42,)`.

Why it matters

- **Integrity:** coordinates, RGB colours, database rows, return bundles can't change by accident.
- **Hashable** (if all items are) → dict key or set element: `{(59.9, 10.7): "Oslo"}`.
- **Lighter:** fixed size, no spare capacity — a 3-item tuple is 72 bytes vs 88 for a list on this build; CPython can build constant tuples at compile time.
- **Thread safety by construction:** an unmutable object can be shared between threads without locks (the objects inside may still be mutable).

How it works — define, access, "update", loop, operators, slice

```
t = (1, 2, 3, 4, 5)
packed = "apple", "banana", "cherry"          # the commas make the tuple
single, not_a_tuple, empty = (42,), (42), ()
print(t, packed, type(single).__name__, type(not_a_tuple).__name__, empty)
print(t[0], t[2], t[-1], t[1:4], t[::-1])
try:
    t[1] = 20
except TypeError as e:
    print("TypeError:", e)
as_list = list(t); as_list[1] = 20; t2 = tuple(as_list)   # "update" = build a new tuple
print("updated copy:", t2)
for item in t:
    print(item, end=" ")
a, b = (1, 2, 3), (4, 5, 6)
print(a + b, a * 2, 3 in a, 7 not in a, (1, 2) < (1, 3))
```

Output:

```
(1, 2, 3, 4, 5) ('apple', 'banana', 'cherry') tuple int ()
1 3 5 (2, 3, 4) (5, 4, 3, 2, 1)
TypeError: 'tuple' object does not support item assignment
updated copy: (1, 20, 3, 4, 5)
1 2 3 4 5
(1, 2, 3, 4, 5, 6) (1, 2, 3, 1, 2, 3) True True True
```

Operator	Meaning	Example
`+`	Concatenate into a new tuple	`(1, 2) + (3,)` → `(1, 2, 3)`
`*`	Repeat	`(1, 2) * 2` → `(1, 2, 1, 2)`
`in` / `not in`	Membership	`3 in (1, 2, 3)` → `True`
`<`, `==`, …	Lexicographic comparison	`(1, 2) < (1, 3)` → `True`
`[i]`, `[i:j:k]`	Index and slice (slices are new tuples)	`(0,1,2,3)[1:3]` → `(1, 2)`

"Update" = list round-trip, or slicing: `t[:1] + (20,) + t[2:]`.

Immutability is shallow

Slots can't change, but a mutable object **inside** can:

```
t = ([1, 2], "x")
t[0].append(3)
print(t)                  # ([1, 2, 3], 'x')
hash(t)                   # TypeError: unhashable type: 'list'
```

A tuple is hashable only if everything inside is.

Tuples as records

A tuple is a **record** where *position has meaning*: `("Ann", "HR", 31)` = name, department, age. Lists hold **many things of one kind**; tuples hold **one thing with several fields** (*Fluent Python*'s guidance) — hence functions return tuples: `divmod(17, 5)` → `(3, 2)`.

Common mistakes

- `(42)` is the int 42; `type(("a"))` is `str` — write `(42,)`.
- `x = 1,` makes a tuple.
- A tuple of lists isn't hashable, so can't be a dict key.

TL;DR

Commas make tuples; shallowly immutable, hashable if contents are, lighter than lists, ideal fixed-shape records.

5.10 Unpacking tuples (and any iterable)

What it is

Unpacking assigns a sequence's items to several names: `a, b, c = (1, "Hello", 3.14)` — for any iterable. It removes index noise (`row[0]`), and powers multiple return values, swaps and pair loops.

How it works

1. The right side is fully evaluated first.
2. Items bind to names in order; counts must match or `ValueError` ("not enough" / "too many values").
3. **Star-unpacking** (`*name`) collects the rest into a **list** (possibly empty); one star per assignment.
4. Unpacking nests: `for name, (lat, lon) in ...`.

```
a, b, c = (1, "Hello", 3.14)
print(a, b, c)
try:
    w, x, y, z = (1, "Hello", 3.14)
except ValueError as e:
    print("ValueError:", e)
first, *middle, last = (1, 2, 3, 4, 5, 6)
print(first, middle, last)
head, *tail = [10]
print(head, tail)
x, y = 1, 2
x, y = y, x                               # swap
print(x, y)
for name, (lat, lon) in [("Oslo", (59.9, 10.7)), ("Lima", (-12.0, -77.0))]:
    print(f"{name}: {lat}, {lon}")
_, _, ext = "archive.tar.gz".partition(".")   # _ = "don't care"
print(ext)
```

Output:

```
1 Hello 3.14
ValueError: not enough values to unpack (expected 4, got 3)
1 [2, 3, 4, 5] 6
10 []
2 1
Oslo: 59.9, 10.7
Lima: -12.0, -77.0
tar.gz
```

`try/except ValueError` lets a program report bad input (a malformed CSV row) instead of crashing. The swap works because `(2, 1)` is built **before** assignment.

Star in calls and literals (the reverse):

Syntax	Meaning	Example
`f(*seq)`	Spread items as positional arguments	`print(*[1, 2, 3])` → `1 2 3`
`[*a, *b]`	Build a list from several iterables	`[*(1, 2), *[3]]` → `[1, 2, 3]`
`(*a, x)`	Build a tuple	`(*t, 99)`
`def f(*args)`	Collect extra positional arguments into a tuple	Chapter on functions

Common mistakes

- Two stars (`a, *b, *c = ...`) → `SyntaxError`.
- `*middle` is always a **list**, not a tuple.
- Unpacking a dict gives **keys**; use `.items()`.

In the interview

> **30-second answer:** "Unpacking binds items of any iterable to names; the count must match or you get ValueError. `first, *rest = seq` gathers the leftovers into a list. It powers multiple return values, swaps without a temp, and clean loops like `for k, v in d.items()`."

- *Is `a, b = b, a` atomic?* → Build tuple, then assign — no temp, but not thread-atomic (other threads can interleave).
- `*_` *?* → Discards any number of items: `first, *_ = row`.

Self-check

1. `a, *b = "hey"` ? → `a='h'`, `b=['e', 'y']`.
2. `x, y = [1, 2, 3]` ? → `ValueError: too many values to unpack (expected 2)`.

TL;DR

`a, b, c = seq`; counts must match; `*rest` gathers a list; `_` for ignored slots; works in loops and nests.

5.11 Joining tuples

What it is

Combining tuples always creates a **new** tuple.

How it works — five ways

```
parts = [(1, 2), (3, 4), (5, 6)]
out = ()
for p in parts:
    out += p                                  # new tuple every time: O(n^2) overall
print("loop     :", out)
print("+        :", (1, 2, 3) + (4, 5, 6))
print("sum      :", sum(parts, ()))           # also O(n^2): repeated +
print("genexpr  :", tuple(x for p in parts for x in p))
print("chain    :", tuple(itertools.chain.from_iterable(parts)))
```

Output (last line: joining 3,000 two-item tuples):

```
loop     : (1, 2, 3, 4, 5, 6)
+        : (1, 2, 3, 4, 5, 6)
sum      : (1, 2, 3, 4, 5, 6)
genexpr  : (1, 2, 3, 4, 5, 6)
chain    : (1, 2, 3, 4, 5, 6)
3000 tuples: sum 14.55 ms vs chain 0.052 ms
```

`+` is fine for two or three. `+=` loops and `sum(parts, ())` copy everything so far each time — O(total²); the docs recommend `itertools.chain` (and `sum` refuses strings). The flattening generator is O(total); `tuple(itertools.chain.from_iterable(parts))` is the idiomatic linear C solution, ~280× faster here.

Common mistakes

- `(1,) + [2]` → `TypeError`; convert first.
- `t + (4)` fails; `t + (4,)` works.

TL;DR

Few → `+`. Many → `tuple(chain.from_iterable(...))` or a flattening generator. Avoid `+=` loops and `sum`.

5.12 Tuple methods

Only two — anything that would modify is missing by design.

Method	Returns	Cost
`count(x)`	Items equal to `x`	O(n)
`index(x[, start[, end]])`	First position of `x`; `ValueError` if absent	O(n)

```
fruit = ("apple", "banana", "cherry", "apple", "cherry")
print(fruit.count("apple"), fruit.count("cherry"), fruit.index("banana"), fruit.index("cherry"))
```

Output:

```
2 2 1 2
```

`index("cherry")` returns the **first** (2). Built-ins `len`, `min`, `max`, `sum`, `sorted` (returns a list), `reversed`, `any`, `all` work on tuples.

TL;DR

`count` and `index`, both O(n); the rest comes from built-ins.

5.13 `namedtuple` and `typing.NamedTuple` — tuples with field names

What it is

`collections.namedtuple` creates a **tuple subclass** whose positions also have names: `p.x` instead of `p[0]`. `typing.NamedTuple` is the class syntax with type hints and defaults.

The problem it solves

`emp[2]` says nothing; `emp.salary` says everything. Named tuples keep every tuple benefit (immutable, hashable, unpackable, tuple-sized — no per-instance `__dict__`, the attribute dictionary ordinary objects carry) while being self-documenting.

How it works

```
Point = namedtuple("Point", ["x", "y"])
p = Point(3, 4)
print(p, p.x, p[1], p._asdict(), p._replace(x=10))
x, y = p                                    # still a tuple

class Employee(NamedTuple):                 # typed, class-style syntax
    name: str
    dept: str
    salary: int = 0
e = Employee("Ann", "HR")
print(e, e.salary)
```

Output:

```
Point(x=3, y=4) 3 4 {'x': 3, 'y': 4} Point(x=10, y=4)
Employee(name='Ann', dept='HR', salary=0) 0
```

Extras: `_asdict()` (to dict, e.g. JSON), `_replace(**changes)` (modified **copy** — the way to "change" a field), `_fields`, `_make(iterable)`.

Trade-offs

	tuple	namedtuple / NamedTuple	@dataclass (Chapter 10)	dict
Access by name	No	Yes	Yes	Yes (`d["k"]`)
Immutable	Yes	Yes	Optional (`frozen=True`)	No
Hashable	If items are	If items are	If frozen	No
Memory	Smallest	Same as tuple	Bigger (`slots=True` shrinks)	Biggest
Methods / validation	No	Limited	Full class	No
Also behaves as a sequence	Yes	Yes (surprise: `p == (3, 4)` is `True`)	No	No

When not to use it

Records needing mutation, validation or methods — or where equality with plain tuples would be a bug — use `@dataclass`.

Real-world examples

`urllib.parse.urlparse()` → `ParseResult(scheme=..., netloc=...)` (a namedtuple subclass). `sys.version_info`, `time.struct_time`, `os.stat_result` are *struct sequences* — the C-level cousin. psycopg 3 (a PostgreSQL driver) offers a `namedtuple_row` row factory.

In the interview

> **30-second answer:** "A namedtuple is a tuple subclass with named fields — immutable, lightweight, unpackable, with `_asdict` and `_replace`. I use it (or `typing.NamedTuple` for type hints) for simple immutable records and switch to dataclasses when I need mutability, validation or methods."

TL;DR

Named fields on a tuple for immutable records; dataclasses for richer objects.

5.14 Choosing: `list` vs `tuple` vs `array.array` vs NumPy

What it is

NumPy's `ndarray` (third-party, `pip install numpy`) is a typed, contiguous, possibly multi-dimensional array whose operations run as compiled loops over the whole array (**vectorisation**).

How they compare — measured

One million integers, "multiply every element by 2":

```
container         memory    x*2 time
list              36.0MB      20.8ms
array.array        8.2MB      45.7ms
numpy              8.0MB       0.6ms
```

`list` holds pointers + int objects (most memory, interpreter loop); `array.array('q')` holds raw 8-byte ints but every read creates a Python int, so a Python loop is **slower** than the list; NumPy holds raw ints **and** does the arithmetic in C → ~35× faster than the list.

Trade-offs

	`list`	`tuple`	`array.array`	`numpy.ndarray`
Mutable	Yes	No	Yes	Yes (fixed size)
Element types	Any, mixed	Any, mixed	One C numeric type	One dtype (element data type: numeric, bool, fixed strings, objects)
Memory for numbers	High	High	Low	Low
Hashable	No	Yes (if items are)	No	No
Fast element-wise maths	No	No	No	Yes (vectorised)
Multi-dimensional	Via nesting	Via nesting	No	Yes
Standard library	Yes	Yes	Yes	No (third-party)
`+` means	Concatenate	Concatenate	Concatenate	**Element-wise add**

When to use which

`list` for changing collections; **`tuple`** for fixed records, dict keys, return values; **`array.array`** for compact numbers with no extra dependency and binary formats; **NumPy** for numerical computing (statistics, images, ML features, linear algebra) — the foundation under pandas, scikit-learn, SciPy and matplotlib.

Common mistakes

- Python-level loops over NumPy arrays lose the speed — use `arr * 2`, `arr[arr > 0]`.

- `np_a + np_b` adds element-wise; concatenate with `np.concatenate`.
- Picking `array.array` for speed — it's for memory.

In the interview

> **30-second answer:** "List for general mutable data, tuple for immutable records and dict keys, `array.array` for compact homogeneous numbers without dependencies, NumPy for fast numerical work because it stores raw values contiguously and runs vectorised C loops."

TL;DR

General → list. Fixed → tuple. Compact numbers → array. Fast numbers → NumPy.

Chapter TL;DR

Topic	Remember
`array` module	Typed, compact, mutable; type codes set size/range; no `sort` / `copy` methods; `clear()` from 3.13; memory win, not speed win.
Lists	Ordered, mutable, mixed types; index/slice/update; `+` new, `extend` in place.
Internals	Dynamic array of pointers; ~12.5% over-allocation → amortised O(1) append; front/middle insert/delete O(n); `in` O(n).
Add/remove	`append`, `extend`, `insert`; `remove` (value), `pop` (index, returns), `del`, `clear`.
Slicing	`[start:stop:step]`; slice assignment can resize; stepped assignment needs equal length.
Comprehensions	`[expr for x in it if cond]`; generator expression for one-pass use; loops for complex logic.
Sorting	`sort()` in place → None; `sorted()` new list; `key`, `reverse`; Timsort is stable, O(n log n), O(n) on sorted input.
Copying	`=` aliases; `copy()` / `[:]` shallow; `deepcopy` recursive; never `[[0]*n]*m`; no mutable defaults.
Tuples	Commas make them; `(x,)` for one; shallowly immutable; hashable if contents are; records.
Unpacking	Counts must match; `*rest` → list; swap with `a, b = b, a`.
Joining tuples	`+` for few; `chain.from_iterable` for many; `sum` / `+=` loops are quadratic.
Tuple methods	`count`, `index`.
namedtuple	Named, immutable records; `_asdict`, `_replace`; dataclass for more.
Choosing	list / tuple / array / NumPy by mutability, memory, and speed needs.